



Antioxidant properties of plum concentrates and powder in precooked roast beef to reduce lipid oxidation

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ABSTRACT

Boneless beef roasts (*Semimembranosus* + *Adductor*) were injected (20%) with a brine containing (1) no plum ingredient (control), (2) 2.5 or 5% fresh plum juice concentrate (FP), (3) 2.5 or 5% dried plum juice concentrate (DP), or (4) 2.5 or 5% spray dried plum powder (PP). Whole roasts were cooked, vacuum-packaged and stored at <4.0 °C for 10 wk. At 2 wk intervals, evaluations were performed on sliced product to determine vacuum-packaged purge, Allo-Kramer shear force, lipid oxidation (TBARS), color space values, and sensory attributes. All plum ingredients reduced TBARS values and had minimal effects on tenderness, sensory characteristics, color and appearance. Small changes in purge, color values, TBARS and some sensory properties were found during storage. These results indicate that 2.5% FP or DP could be incorporated into precooked beef roasts to reduce lipid oxidation and potentially, warmed-over flavor (WOF).

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1. Introduction

Lipid oxidation can have negative effects on the quality of meat and meat products causing changes in sensory attributes (color, texture, odor, and flavor) and nutritional quality. Oxidation of unsaturated lipids in cooked meat during storage and reheating leads to the formation of stale or rancid flavors termed as warmed-over flavor (WOF) (Sato & Hegarty, 1971). WOF in cooked meats has become more of a problem in recent years due to the increase in precooked meats at food service facilities (Jayasingh & Cornforth, 2003).

Synthetic antioxidants such as butylated hydroxyanisole (BHA) and butylated hydroxytoluene (BHT) have been widely used in meat to suppress or retard the development of WOF (Chastain, Huffman, Hsieh, & Cordray, 1982; Chen, Pearson, Gray, & Merkel, 1984; Sato & Hegarty, 1971). However, the use of synthetic antioxidants has come under more scrutiny due to their potential toxicological effects (Decker & Mei, 1996; Rababah, Hettiarachchy, & Horax, 2004). Consumer preferences for “natural” products have resulted in increased interest in the use of natural antioxidants from rosemary, sage, aloe vera, mustard, tea catechins, whey protein concentrate, and cottonseed meals (McCarthy, Kerry, Lynch, & Buckley, 2001; Rhee, Ziprin, & Calhoun, 2001).

Scientists have demonstrated that some fruits and berries are a good source of antioxidants due to their flavonoid and polyphenol content (Skrede & Wrolstad, 2002; Wang, Cao, & Prior, 1996). Plum-derived food ingredients have been reported to function as antioxidants, antimicrobials, fat replacers, and flavorants. Hamburgers containing dried plum puree have been reported to retain 15.8% more moisture when reheated and held for up to 4 h (Decker, 1999). Moreover, dried plum puree at 3% and higher has been shown to be as effective as BHA/BHT in retarding lipid oxidation in precooked pork patties (Nuñez de Gonzalez, Boleman, Miller, Keeton, & Rhee, 2008). Lee and Ahn (2005) likewise observed that plum extract used at >2% in turkey breast rolls and irradiated at 3.0 kGy was effective at retarding lipid oxidation while enhancing juiciness. These studies indicate that dried or fresh plum products might also serve to protect flavor in precooked, whole muscle meats injected with a brine marinade. Therefore, the objective of this study was to determine the physical, chemical, and sensory attributes of brine-injected, cooked roast beef containing varying levels of fresh and dried plum ingredients and subjected to refrigerated storage over a 10 wk period.

2. Materials and methods

2.1. Roast beef processing

Fresh, boneless, USDA-Select top rounds (*Semimembranosus* + *Adductor*), IMPS # 169A, were purchased from a local food

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service purveyor (3–4 d postmortem) in College Station, Texas. Whole, boneless top rounds were trimmed of excess of fat and connective tissue and then split into equal halves to form roasts weighing 1.8–3.6 kg. A total of 168 rounds (Table 1) or 24 rounds per treatment were randomized and injected (Inject Star, Model BI-72, Brookfield, CT) with a brine solution (20% by weight of raw product) to contain (1) no plum ingredient (control), (2) 2.5% or 5% fresh plum juice concentrate (FP), (3) 2.5% or 5% dried plum juice concentrate (DP), or (4) 2.5% or 5% spray dried plum powder (PP). All plum ingredients were obtained through the California Plum Board (Sunsweet Growers Inc., Yuba City, CA). The other ingredients, as percent after injection into the product, included salt (1.7%, Salt Culinox 999 food grade, Morton International Inc., Chicago, IL), dextrose (0.9%, Newly Weds® Foods, Bedford Park, IL), alkaline phosphate (0.35%, Bifisol® 512, BK Giulini Co., Simi Valley, CA), potassium lactate (3%, Purasal P HiPure 60, Purac America Inc., Lincolnshire, IL), and water. Beef roasts were prepared in a state inspected (Texas Department of Health), commercial-scale pilot plant located in the Rosenthal Meat Science and Technology Center at Texas A&M University.

Following injection, the rounds were allowed to drain to remove excess brine and then re-weighed, and the pumped weights recorded to determine the percent pump. After brine injection, rounds within each treatment and injection level were vacuum tumbled (Model VT500, Leland Southwest Inc., Ft. Worth, TX; 846.6 mbar, 4–5 rpm) for 1 h to aid brine ingredient distribution and absorption. Tumbling was performed in a room maintained at $\leq 5^\circ\text{C}$. Following tumbling, product weights were recorded. Then, rounds were vacuum-packaged in Cryovac® cook-in bags (type CN530, product # 97725, 30.48 cm by 60.96 cm, Cryovac North America, Duncan, SC) and re-weighed and packaged weights were recorded.

Prior to cooking, rounds were stored overnight in a cooler maintained at $<4^\circ\text{C}$ to allow for brine equalization. An automated, sequential processing cycle for beef roasts was selected to achieve a product endpoint temperature of 62.8°C in a commercial processing smokehouse (Model 1000, Alkar, DEC International Inc., Lodi, WI). The smokehouse was set for the following conditions: 1 h at 60°C , 1 h at 65.6°C , 1 h at 68.3°C , 1 h at 71.1°C and 2–4 h at 73.8°C . Upon completion of the cooking cycle, the whole roasts were cooled to 37.7°C by showering with cold water for 6–8 min. Then, the products were chilled according to the USDA-FSIS (1999, Appendix B) guidelines and stored in cardboard boxes 2–3 d in the dark at $<4^\circ\text{C}$.

2.2. Cook loss

Chilled roasts were weighed prior to opening the cook-in bags to obtain cooked, packaged weights. Individual roasts were then

removed from the package, drained thoroughly, and the cooked meat weights and empty package weights recorded to calculate total cook loss as follows:

$$\text{Cook loss (\%)} = \left[\frac{\text{purge weight}}{\text{cooked roast weight} + \text{purge weight}} \right] \times 100$$

After determining cook loss, roasts were assigned to storage treatments of 0, 2, 4, 6, 8, and 10 wk, vacuum-packaged (Ultravac® 2100, vacuum setting 7 and seal setting 7, KOCH Inc., Kansas City, MO) in Cryovac® bags (type BH620T, product # 9D640, 33.02 cm by 66.04 cm, Cryovac North America, Duncan, SC) and stored in a cooler maintained at $<4^\circ\text{C}$. At the end of each designated storage period, samples were taken for Allo-Kramer shear force, descriptive attribute sensory panel evaluation, color space (L^* , a^* , and b^*) values, lipid oxidation, and proximate analysis. An additional set of sliced samples was stored 21 d and then assessed for vacuum purge at the end of the 21 d (Fig. 1). All sample slices were vacuum-packaged in Cryovac® bags (type B540, product # 90184, 17.78 cm by 30.48 cm, Cryovac North America, Duncan, SC) and stored at $<4^\circ\text{C}$ until analyses could be performed the following day.

2.3. Experimental design

The experiment was designed as a complete randomized factorial design (7 ingredient treatments $\times$ 6 storage times). Ingredient treatments included a control (no antioxidant), 2.5% and 5% FP, 2.5% and 5% DP, and 2.5% and 5% PP. Eighty-four top rounds were split along the longitudinal axis into equal halves and randomly assigned to the treatments and storage combination (Table 1). Rounds were cooked and then stored at $<4^\circ\text{C}$ for 0, 2, 4, 6, 8 and 10 wk. At 2 wk intervals samples were assessed for vacuum-packaged purge, Allo-Kramer shear force, lipid oxidation, color space values, sensory attributes and sensory color over a period of 10 wk. Proximate analysis was performed on finished products from each ingredient treatment.

2.4. Sliced vacuum-package purge at 21 d post whole roast storage

At the end of each designated storage period (0, 2, 4, 6, 8, 10 wk), a 0.454 kg sample of 0.31 cm thick slices from each roast was vacuum-packaged and stored in the dark for an additional 21 d at $<4^\circ\text{C}$. After 3 wk, each package was weighed to obtain a total package weight. Slices were then removed from the package, patted dry with a towel to remove excess moisture, and reweighed. Purge was calculated as total package weight minus package and sample weights. Percent vacuum-package purge was determined using the following formula:

$$\text{Vacuum package purge (\%)} = \left[\frac{\text{purge weight}}{\text{sample weight} + \text{purge weight}} \right] \times 100$$

2.5. Color evaluations

L^* , a^* , b^* color space values for roast treatments at each designated storage time were obtained by reflectance using a Minolta Colorimeter (Model CR-300, Minolta Co., Ramsey, NJ), with an 8 mm viewing port and illuminant D_{65} , calibrated to a white tile standard surface ($L^* = 97.55$, $a^* = -0.02$, $b^* = 1.56$ at channel 00). The colorimeter port was covered with clear Saran® Wrap and then calibrated. Color measurements were made on each slice at three randomly selected locations. The results were expressed as positive L^* (lightness), a^* (redness) and b^* (yellowness) color space values.

Table 1
Experimental design of roast beef allocation by treatment and storage time

Treatment ^a	Storage time (week)						Total
	0	2	4	6	8	10	
Control	4	4	4	4	4	4	24
FP 2.5%	4	4	4	4	4	4	24
FP 5%	4	4	4	4	4	4	24
DP 2.5%	4	4	4	4	4	4	24
DP 5%	4	4	4	4	4	4	24
PP 2.5%	4	4	4	4	4	4	24
PP 5%	4	4	4	4	4	4	24
Total							168

^a Control: no plum ingredient; FP: fresh plum juice concentrate; DP: dried plum juice concentrate; PP: spray dried plum powder.

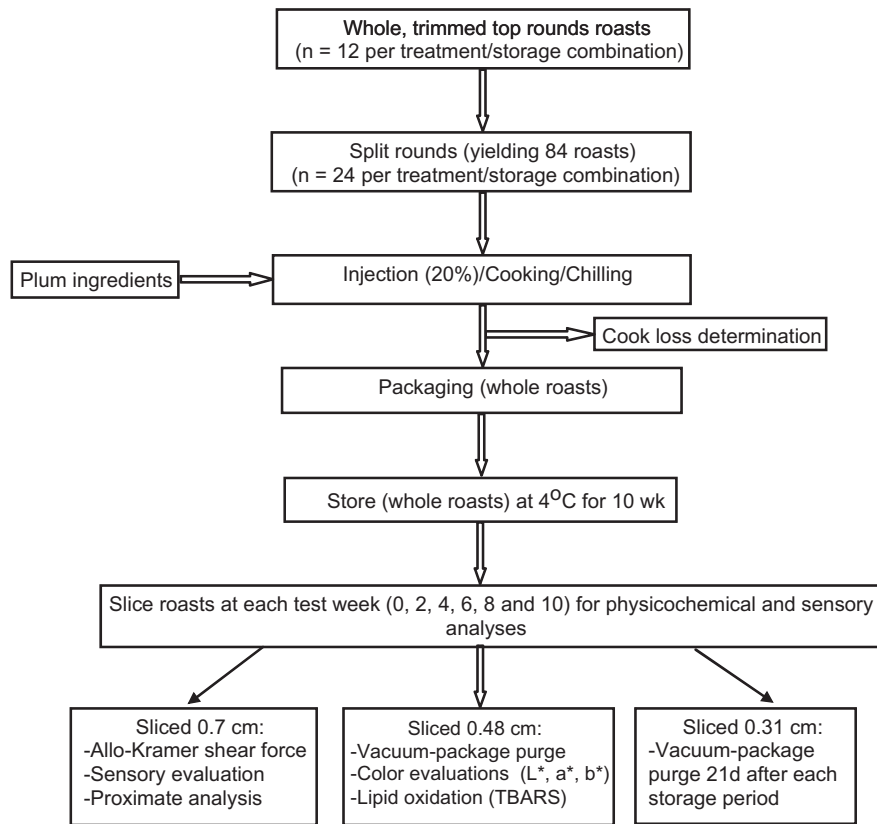


Fig. 1. Protocols for roast beef processing, and physicochemical and sensory analyses.

2.6. Lipid oxidation

2-Thiobarbituric acid reactive substances (TBARS) were determined for each treatment at each designated storage time using the TBA distillation procedure of [Tarladgis, Watts, Younathan, and Dugan \(1960\)](#) as modified by [Rhee \(1978\)](#). Each sample was analyzed in duplicate and results were expressed as mg of malonaldehyde (MDA) per kg of meat.

2.7. Allo-Kramer shear force

Allo-Kramer shear force measurements were performed on each treatment at the end of each designated storage period using an Instron Universal Testing Machine (Model 1011, Instron Co., Houston, TX). A standardized specimen size (2 by 5 cm²) was cut, weighed and placed flat in a 10-blade Allo-Kramer shear cell attached to a 500 kg load cell with a standard load range setting of 100 kg. Kilograms of shear force were recorded and divided by the sample weight to determine the shear force in kg/g of sample. This value was then converted to Newtons/g by multiplying kg × 9.8.

2.8. Proximate analysis

Percentages of moisture, fat, and protein were determined on finished products according to [AOAC \(2000\)](#) methods. Slices from roasts assigned to each treatment at 0 wk storage period were homogenized in a food processor (Model R6, Robot Coupe S.A., Bagnolet, France) before sampling. Percentages of moisture and fat were determined using a convection air-dry oven (AOAC 950.46) and Soxhlet ether extraction (AOAC 985.15) methods, respectively. Crude protein percentage was determined by Dumas combustion method (AOAC 992.15) for gaseous N₂ using a Leco FP-

528 Protein Analyzer (St. Joseph, MO). The instrument was standardized using oatmeal (Part No 502-276, %N = 9.56 ± 0.04) after each block of 10–12 samples. Crude protein percentage was calculated as 6.25 times the percent nitrogen. All analyses of the samples were performed in duplicate.

2.9. Sensory evaluation

A seven-member expert, descriptive attribute sensory panel in the Texas A&M University Sensory Testing Facility was used to evaluate roast samples for aromatics, flavor, feeling factors, basic tastes, and texture attributes. The panel had been selected and trained as described by [AMSA \(1995\)](#), [Cross, Moen, and Stanfield \(1978\)](#), and [Meilgaard, Civille, and Carr \(1999\)](#). All panelists had more than five years of experience in Spectrum™ descriptive flavor and texture analysis ([Meilgaard et al., 1999](#)). Panelists underwent ballot development and training sessions using beef roasts prepared without antioxidant (control “as is”) and with antioxidant treatments (2.5% or 5% FP, 2.5% or 5% DP and 2.5% or 5% PP). Six training sessions were conducted prior to testing. Panelists underwent performance evaluation as specified in the guidelines developed by [AMSA \(1995\)](#) prior to initiation of the study to assure panelists were sufficiently trained. Beef roast samples were evaluated for aromatics (cooked beef/brothy, cooked beef fat, chemical taste, serum/bloody, and plum/prune); feeling factors (astringent, metallic, and chemical burn); basic tastes (salt, sour, bitter, and sweet); and texture attributes (springiness, juiciness, hardness, cohesiveness, and denseness) as defined in [Table 2](#). All samples were scored using the 0–15 Spectrum™ Universal intensity scale ([Meilgaard et al., 1999](#)), where 0 = absence of an attribute and 15 = extremely intense. In addition, panelists evaluated texture (springiness, juiciness, hardness, cohesiveness and denseness) using the 0–15 Spectrum™ Universal Intensity scale ([Meilgaard](#)

Table 2

Aromatics, feeling factors, basic tastes and texture attribute definitions, references and intensity based on a 15-point scale and established after consensus discussions with the expert panel

Attribute	Definition	Reference products and intensity
<i>Aromatics</i>		
Cooked beef/brothy	Aromatic associated with cooked beef muscle meat and broth/broth like	Concentrated beef broth (Swanson® beef broth, Campbell Soup Co., Camden, NJ, USA) = 15
Cooked beef fat	Aromatic associated with cooked beef fat	Cooked beef fat = 15
Chemical taste	Aromatic associated with many different type of compounds such as solvents, cleaning compounds and hydrocarbons	Cherry juice with 500 ppm bleached added (odor only) = 15
Serum/bloody	Aromatic associated with raw beef lean	Tenderloin steak cooked to rare degree of doneness = 6.5
Prune/plum	Aromatic associated with sweet, fruity aromatic of plum or a browned aroma, reminiscent of prunes	Prunes placed in mouth = 15
<i>Feeling factors</i>		
Astringent	The shrinking or puckering of the tongue surface caused by substances such as tannins or alum	Cooked 1.27 cm sample of liver = 15
Metallic	A feeling factor on the tongue stimulated by metal or the aromatics associated with metals	Black tea (Lipton, Englewood Cliffs, NJ, USA) bag placed in hot water and allowed to sit for 1 h = 6.5
Chemical burn	Chemical feeling factor associated with high concentrations of irritants to the mucous membrane of the oral cavity	White vinegar (HJ. Heinz Co., Pittsburg, PA, USA) = 15
<i>Basic tastes</i>		
Salt	Basic taste stimulated by sodium salts	Potato chips (Pringles, Procter & Gamble, Cincinnati, OH, USA) = 13
Sour	Basic taste stimulated by acids	Lemon juice (Real Lemon®, Mott's Inc., Stanford, CT, USA) = 8.0
Bitter	Basic taste stimulated by bitter agents such as caffeine, quinine, etc	Caffeine (0.15% solution) = 10.0
Sweet	Basic taste stimulated by sugars	Chocolate bar (The Hershey Co., PA, USA) = 10.0
<i>Textures</i>		
Springiness	Degree to which sample returns to original shape after a certain time period	Marshmallow (miniature, Wal-Mart Inc. AR, USA) = 9.5
Juiciness	Amount of wetness/juiciness released from sample	Apples (Red Delicious) = 10.0
Hardness	Force required to bite through sample	Hard candy (Life Savers®, WM. Wrigley JR. Co., Chicago, IL, USA) = 14.5
Cohesiveness	The amount the sample deforms rather than shears/cuts	Candy chews (Starburst, Master-foods USA®, Hackettstown, NJ, USA) = 12.5
Denseness	Compactness of the cross section	Malted milk balls (Whopper, The Hershey Co., PA, USA) = 6.0

et al., 1999), where 0 not springy, dry, soft, crumbles, airy and 15 = very springy, juicy, hard, defined particle size, dense for each attribute.

Samples of refrigerated roasts from each treatment were evaluated at 2 wk intervals. On each testing day, seven samples were evaluated per day during one session. The order of the treatments was randomized and a warm-up was presented to judges before sample evaluation to ensure they were familiar with the treatment attributes to be tested. The stimuli used for warm-up were roasts from a control formulation. The samples were coded with random

three digit numbers to mask the treatment identity and placed in clear 6-ounce samples cups with lids. Panelists received two randomly selected samples (2.54 cm²) from each roast maintained at 4–6 °C. Panelists evaluated samples in isolated booths fitted with a breadbox server and red filtered, incandescent lighting to mask color differences. Distilled water at room temperature, unsalted crackers and ricotta cheese were given to judges to cleanse their mouths.

In addition to flavor and texture, the trained panelists also evaluated roast samples for subjective color. Slices from each treatment were selected at random and displayed for evaluation of color intensity, greyness, off-color, and iridescence. Panelists scored roast slices for overall color using an 8-point descriptive scale (1 = grey; 8 = dark reddish pink). Using a 6-point descriptive scale, panelists also scored percent surface discoloration (grey and/or brown) (1 = no/0% surface grey color, no/0% surface brown color; 6 = total/100% surface grey color, total/100% surface brown color and iridescence (1 = none, 0%; 6 = very strong, 100%). All samples were evaluated under standard room fluorescent lighting (Econ-o-watt F40CW/RS/EW, 34W, Philips Electronics North America, New York, NY). After color evaluation by the sensory panel, *L*^{*} (lightness), *a*^{*} (redness), and *b*^{*} (yellowness) color space values were determined on the same roast slices using a Minolta colorimeter (Model CR-300, Minolta Co., Ramsey, NJ).

2.10. Statistical analysis

Data were analyzed as a 7 (ingredient treatments) by 6 (storage times) factorial design using the general linear model (GLM) procedure of the statistical analysis system (SAS® Institute, 1995). Ingredient treatment (no antioxidant, 2.5% FP, 5% FP, 2.5% DP, 5% DP, 2.5% PP or 5% PP) and storage time (0, 2, 4, 6, 8, or 10 wk) were the main effects for the data. Analysis of variance was used to determine statistical differences among the main effects and their interactions at a significance level of *P* < 0.05. Least squares means were achieved using the PDIFF statement of GLM (SAS, 1995) to identify significant treatment effects. Descriptive attribute sensory evaluation data were pooled across panelists and were analyzed as previously described.

3. Results and discussion

3.1. Physical and chemical properties of cooked roast beef with fresh/dried plum ingredients

Results for cook loss, vacuum-package purge, sliced vacuum-package purge at 21 d post storage, objective color, lipid oxidation, and shear force values of boneless, precooked, beef roasts are presented in Table 3. Cook loss percentages for roasts containing FP at 2.5% and 5% or DP at 2.5% were not significantly different from those of the control. However, cook loss was 19.5%, 19.8% and 27% more for DP 5%, PP 2.5%, and PP 5%, respectively, when compared to the control at 16.6%. No differences in cook loss percentages were noted among beef roasts containing FP 2.5%, DP 5% or PP 2.5%. With increasing levels of plum material, percent cook loss increased for roasts injected with DP and PP. Overall, percent cook loss of the 2.5% and 5% FP and 2.5% DP treatments would be comparable to conventionally processed beef roast. The higher percent cook loss found for DP 5% and both PP treatments was caused by the product's inability to bind water during cooking.

Most vacuum-packaged whole roasts showed little or no change in percent purge loss during refrigerated storage except for roasts injected with 2.5% DP which had slightly (*P* < 0.05) more purge loss than all other treatments. Cook loss was similar to the

Table 3

Least squares means for cook loss, vacuum-package purge, sliced vacuum-package purge at 21 d post storage, L^* a^* b^* color space values, TBARS values, and Allo-Kramer shear force values of boneless roast beef with fresh/dried plum ingredients

Treatment ^e	Cook loss (%)	Vacuum-package purge (%)	Sliced vacuum-package purge at 21 d post storage (%)	Color space values			TBARS (mg MDA/kg)	Allo-Kramer shear force value (N/g sample)
				L^*	a^*	b^*		
Control	16.6 ^{ab}	3.0 ^b	2.6 ^b	54.5 ^c	10.2 ^b	12.2 ^{cd}	0.62 ^c	65.7 ^{abc}
FP 2.5%	18.2 ^{bc}	2.5 ^{ab}	2.5 ^b	52.5 ^b	9.5 ^a	11.3 ^{ab}	0.26 ^{ab}	76.4 ^d
FP 5%	16.8 ^{ab}	2.6 ^{ab}	2.7 ^b	50.8 ^a	9.4 ^a	11.1 ^{ab}	0.16 ^a	72.5 ^{cd}
DP 2.5%	15.2 ^a	4.0 ^c	2.4 ^b	52.7 ^b	12.2 ^c	10.9 ^{ab}	0.39 ^b	62.7 ^a
DP 5%	19.5 ^c	3.0 ^b	2.9 ^b	50.5 ^a	11.8 ^c	10.7 ^a	0.31 ^{ab}	70.6 ^{bcd}
PP 2.5%	19.8 ^c	2.5 ^{ab}	2.7 ^b	52.7 ^b	10.2 ^b	12.5 ^d	0.40 ^b	65.7 ^{ab}
PP 5%	27.7 ^d	2.2 ^a	1.7 ^a	53.6 ^{bc}	10.3 ^b	11.6 ^{bc}	0.33 ^{ab}	71.5 ^{bcd}
SEM ^f	0.7	0.3	0.2	0.4	0.2	0.3	0.1	1.96

^{abcd} Least squares means in the same column without a common superscript letter differ ($P < 0.05$).

^e Control: no plum ingredient; FP: fresh plum juice concentrate; DP: dried plum juice concentrate; PP: spray dried plum powder.

^f SEM: standard error of the mean.

control after cooking, but under vacuum storage, approximately 1% more moisture than the control was slowly released from the tissue matrix. Roasts with 5% PP had the lowest vacuum-package purge, but this was likely due to less available moisture in the product after cooking.

Beef roasts with fresh/dried plum ingredients had slightly ($P < 0.05$) lower L^* values and were darker, with the exception of the 5% PP treatment, which was equivalent to the control. Increasing the level of FP or DP to 5% also decreased L^* color space values and made the samples slightly ($P < 0.05$) darker than the 2.5% level. Similarly, Lee and Ahn (2005) reported that L^* values decreased in turkey breast rolls with 3% plum puree extract. Samples injected with FP were slightly less red while roasts with DP were slightly more red, when compared to the control. PP was not different from the control in redness values. The a^* color space values indicated that addition 2.5% or 5% FP could decrease ($P < 0.05$) redness slightly while inclusion of 2.5% or 5% DP could increase redness slightly. In addition to changes in redness, FP and DP decreased yellowness (b^* color space values) in roast beef samples, but PP treatments were similar to the control for b^* value.

The addition of fresh and dried plum ingredients significantly decreased TBARS values when compared to the control (Table 3). For roasts containing plum ingredients, increasing the level of ingredient added to the brine tended to decrease TBARS values. Overall, the addition of plum ingredients significantly reduced lipid oxidation and the potential for WOF development in beef roasts. Previous studies also have indicated dried plum puree to be an effective antioxidant in meat products (Kreuzer, 2001; Movileanu, 2003; Nuñez de Gonzalez et al., 2008). The principal phytochemicals in dried plums are phenolic compounds that include phenolic acid derivatives, flavonoids and coumarins (Macheix, Fleuriet, & Billot, 1990). In addition, dried plums contain large amounts of compounds previously shown to have antioxidant activity such as neochlorogenic acid, chlorogenic acid (Donovan, Meyer, & Waterhouse, 1998; Raynal, Moutounet, & Souquet, 1989) and cryptochlorogenic acid (Nakatani et al., 2000). Chlorogenic acid isomers have high antioxidant activities and inhibit low density lipoprotein (LDL) oxidation (Meyer, Donovan, Pearson, Waterhouse, & Frankel, 1998). By scavenging peroxy radicals, chlorogenic acid has been shown to inhibit the initiation of chain lipid peroxidations by organic free radical (Kono et al., 1997).

Only small differences in total shear value were observed as a result of the inclusion of fresh and dried plum ingredients. The 2.5% FP treatment required slightly ($P < 0.05$) less force to shear, but the magnitude of this difference would not likely be detected by consumers.

With changes in percent cook loss, corresponding changes in the percentages of moisture, fat, and protein would be expected (Table 4). Small changes in percent moisture were evident and

Table 4

Least squares means for moisture, fat, and protein of boneless roast beef with fresh/dried plum ingredients

Treatment ^g	Moisture (%)	Fat (%)	Protein (%)
Control	68.2 ^d	1.9 ^c	25.1 ^c
FP 2.5%	68.2 ^d	1.2 ^a	27.0 ^e
FP 5%	67.4 ^{bc}	1.5 ^b	24.5 ^b
DP 2.5%	68.7 ^e	1.2 ^a	24.1 ^a
DP 5%	67.1 ^b	1.8 ^c	25.4 ^{cd}
PP 2.5%	67.6 ^c	2.3 ^e	25.6 ^d
PP 5%	64.9 ^a	2.1 ^d	27.8 ^f
SEM ^h	0.1	0.1	0.1

^{abcdef} Least squares means in the same column without a common superscript letter differ ($P < 0.05$).

^g Control: no plum ingredient; FP: fresh plum juice concentrate; DP: dried plum juice concentrate; PP: spray dried plum powder.

^h SEM: standard error of the mean.

reflected in the cooking losses. FP and DP at 5% and both PP treatments were lower in percent moisture as compared to the control. Conversely, roast beef samples containing PP had a higher percentage of fat and protein as compared to control samples due to the moisture loss. Inclusion of FP at 2.5%, 5% and 2.5% DP resulted in small declines in fat content. Percent protein varied among the FP and DP treatments, which reflects the meat dilution effect of the added ingredient (5% FP) or a concentration effect caused by loss of moisture (5% PP).

3.2. Storage effects on physical and chemical properties of precooked roast beef

Values for percent vacuum-package purge, percent sliced vacuum-package purge at 21 d post storage, objective color, lipid oxidation, and shear force for boneless beef roasts during 10 wk of refrigerated vacuum-package storage are shown in Table 5. Percent cook loss did not change due to storage. However, the percentage of purge from whole roasts increased after 2–4 wk, and remained relatively constant thereafter through wk 10. For sliced samples stored 21 d beyond each designated storage week, percent sliced vacuum-package purge was less ($P < 0.05$) after 4 wk and beyond.

Objective L^* and b^* color space values were variable during refrigerated vacuum-package storage since all treatments were pooled over storage. In general, a^* (redness) values decreased ($P < 0.05$) in whole beef roasts stored under refrigeration for 10 wk. Overall, a slight increase ($P < 0.05$) in TBARS values was noted from 6 to 10 wk when compared to day 0 indicating that some lipid oxidation occurred during storage. Because the TBARS values in Table 5 are based on pooled data (pooled over fresh and dried plum treatments), they do not reflect the beneficial antioxidant effects

Table 5
Least squares means for vacuum-package purge, sliced vacuum-package purge at 21 d post storage, $L^* a^* b^*$ color space values, TBARS values, and Allo-Kramer shear force values of boneless roast beef stored under refrigerated conditions

Storage week	Vacuum-package purge (%)	Sliced vacuum-package purge at 21 d post storage (%)	Color space values			TBARS (mg MDA/kg)	Allo-Kramer shear force value (Newtons/g sample)
			L^*	a^*	b^*		
0	NT ^e	3.3 ^d	52.3 ^{bc}	11.4 ^c	11.2 ^b	0.20 ^a	75.5 ^c
2	1.9 ^a	2.8 ^{cd}	53.1 ^c	10.7 ^b	11.9 ^b	0.29 ^{ab}	72.5 ^{bc}
4	3.3 ^c	2.5 ^{bc}	51.8 ^{ab}	10.4 ^b	10.4 ^a	0.31 ^{ab}	65.7 ^a
6	2.5 ^{ab}	1.8 ^a	55.3 ^d	9.6 ^a	11.9 ^b	0.51 ^c	67.6 ^{ab}
8	3.3 ^{bc}	2.3 ^{bc}	50.9 ^a	11.3 ^c	13.3 ^c	0.39 ^{bc}	65.7 ^a
10	3.2 ^c	2.2 ^{ab}	51.4 ^{ab}	9.7 ^a	10.1 ^a	0.41 ^{bc}	70.6 ^{abc}
SEM ^f	0.2	0.2	0.4	0.2	0.3	0.1	1.96

^{abcd} Least squares means in the same column without a common superscript letter differ ($P < 0.05$).

^e NT: not tested.

^f SEM: standard error of the mean.

Table 6
Least squares means of descriptive attribute sensory panel scores for aromatics^e, feeling factors^e, basic tastes^e, and textures^f of boneless roast beef with fresh/dried plum ingredients

	Treatments ^g							SEM ^h
	Control	FP 2.5%	FP 5%	DP 2.5%	DP 5%	PP 2.5%	PP 5%	
<i>Aromatics</i>								
Cooked beef/ brothy	5.7	5.6	5.4	5.4	5.6	5.6	5.7	0.3
Cooked beef fat	1.7	1.6	1.7	1.7	1.7	1.5	1.4	0.1
Chemical taste	0.8	0.8	0.9	0.8	0.8	0.8	1.0	0.2
Serum/bloody	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.1
Plum/prune	0.5 ^a	1.5 ^{bc}	2.6 ^d	1.5 ^{bc}	2.2 ^{cd}	0.8 ^{ab}	1.0 ^{ab}	0.3
<i>Feeling factors</i>								
Astringent	2.3	2.5	2.6	2.4	2.3	2.4	2.4	0.1
Metallic	2.0	2.1	2.2	1.9	2.0	1.9	2.0	0.1
Chemical burn	0.7	0.9	1.1	0.9	0.8	0.9	0.8	0.2
<i>Basic Tastes</i>								
Salt	6.5	7.0	6.8	7.4	6.5	7.5	6.5	0.4
Sour	2.3	2.2	2.5	2.4	2.3	2.0	2.2	0.2
Bitter	1.8	1.6	1.9	1.9	1.6	1.7	1.7	0.2
Sweet	1.5 ^{ab}	1.5 ^{ab}	1.8 ^b	1.3 ^a	1.9 ^b	1.3 ^a	1.5 ^{ab}	0.2
<i>Textures</i>								
Springiness	4.7	4.9	5.0	4.9	4.8	5.1	4.8	0.2
Juiciness	4.7	4.8	4.6	4.5	4.6	5.1	4.6	0.2
Hardness	5.5	5.7	5.1	5.5	5.7	5.4	5.4	0.2
Cohesiveness	6.3	6.1	5.9	6.2	6.5	6.0	6.1	0.2
Denseness	6.8	6.6	6.4	6.8	6.7	6.9	6.6	0.3

^{abcd} Least squares means in the same row without a common superscript letter differ ($P < 0.05$).

^e Based on the 0–15 Spectrum™ Universal Intensity scale (0 = absence of flavor; 15 = extremely intense flavor).

^f Based on the 0–15 Spectrum™ Universal Intensity scale (0 = not springy, dry, soft, crumbles, airy; 15 = very springy, juicy, hard, defined particle size, dense).

^g Control: no plum ingredient; FP: fresh plum juice concentrate; DP: dried plum juice concentrate; PP: spray dried plum powder.

^h SEM: standard error of the mean.

of the fresh and dried plum treatments. When data were analyzed according to treatments (Table 3), all plum ingredient treatments reduced lipid oxidation as determined by TBARS content. Allo-Kramer shear force values for precooked beef roasts declined slightly over 8 wk, but at 10 wk they were not different from the control.

3.3. Sensory properties of precooked roast beef

Descriptive attribute sensory panel scores revealed only marginally detectable differences in the plum/prune aromatic and sweet taste among the roasts injected with plum ingredients (Table 6). Plum/prune aromatic, although barely detected, was higher ($P < 0.05$) in roasts containing FP or DP as compared to the control,

Table 7
Least squares means of descriptive attribute sensory panel scores for color^d, grey color^e, off color^f, and iridescence^g of boneless roast beef slices with fresh/dried plum ingredients

Treatment ^h	Color	Grey color	Off color	Iridescence
Control	5.8 ^{bc}	1.9	1.7	2.5
FP 2.5%	4.9 ^a	2.6	1.6	2.3
FP 5%	5.2 ^{ab}	1.9	1.8	2.5
DP 2.5%	5.5 ^{abc}	1.8	1.5	2.5
DP 5%	5.4 ^{abc}	2.0	1.8	2.7
PP 2.5%	6.0 ^c	1.8	1.5	2.1
PP 5%	5.2 ^{ab}	2.0	1.6	2.4
SEM ⁱ	0.3	0.3	0.2	0.2

^{abc} Least squares means in the same column without a common superscript letter differ ($P < 0.05$).

^d Based on a 8-point intensity scale (1 = grey; 8 = dark reddish pink).

^e Based on a 6-point intensity scale (1 = no/0% surface grey color; 6 = total/100% surface grey color).

^f Based on a 6-point intensity scale (1 = no/0% surface brown color; 6 = total/100% surface brown color).

^g Based on a 6-point intensity scale (1 = none, 0%; 6 = very strong, 100%).

^h Control: no plum ingredient; FP: fresh plum juice concentrate; DP: dried plum juice concentrate; PP: spray dried plum powder.

ⁱ SEM: standard error of the mean.

and tended to be higher in the 5% level as compared to the 2.5% level. Roasts injected with PP were similar to the control in plum/prune aromatic intensity. Although significantly different ($P < 0.05$), sweet taste scores were similar among the control and beef roasts injected with plum ingredients.

In addition to the objective color measurements, subjective color observations by the trained panelists were also made. The addition of plum ingredients to roast beef was found to decrease ($P < 0.05$) redness slightly only in the 2.5% FP treatment (Table 7). Other treatments did not differ in redness, surface grey color, off-color (surface browning), or iridescence. Roasts injected with 2.5% PP had slightly ($P < 0.05$) higher color scores than samples containing FP. Thus, injection of plum ingredients into beef roasts had minimal effects on sensory color and appearance attributes.

3.4. Storage effects on sensory profiles of precooked roast beef

Sensory scores for aromatics, feeling factors, basic tastes and texture attributes of beef roasts stored up to 8 wk under refrigerated conditions are presented in Table 8. Beef/brothy and beef fat aromatics increased through wk 4 of storage, but then declined slightly ($P < 0.05$) by wk 8. Other aromatics were only marginally detected (mean scores of 0–1.5 on a 0–15 scale). Astringency (a feeling factor) increased slightly ($P < 0.05$) on wk 4 and 8 while metallic was slightly ($P < 0.05$) higher on wk 8. Chemical burn was only slightly ($P < 0.05$) higher on wk 4 and 8. However, feeling factors although detectable, were very low in magnitude and ran-

Table 8

Least squares means of descriptive attribute sensory panel scores for aromatics^e, feeling factors^e, basic tastes^e, and textures^f of boneless roast beef stored under refrigerated conditions

	Storage week					SEM ^g
	0	2	4	6	8	
<i>Aromatics</i>						
Cooked beef/brothy	3.6 ^a	6.3 ^c	7.0 ^d	6.5 ^{cd}	4.6 ^b	0.2
Cooked beef fat	0.4 ^a	1.6 ^b	2.2 ^d	1.9 ^c	1.9 ^c	0.1
Chemical taste	0.9 ^{ab}	0.9 ^{ab}	1.1 ^b	0.4 ^a	1.1 ^b	0.2
Serum/bloody	0.4 ^c	0.0 ^a	0.1 ^b	0.0 ^a	0.0 ^a	0.1
Plum/prune	1.4 ^{ab}	1.5 ^{bc}	1.5 ^{abc}	0.9 ^a	2.0 ^c	0.2
<i>Feeling factors</i>						
Astringent	2.2 ^a	2.3 ^{ab}	2.7 ^c	2.4 ^{ab}	2.5 ^{bc}	0.1
Metallic	1.9 ^a	2.0 ^{ab}	2.0 ^{ab}	1.9 ^a	2.2 ^b	0.1
Chemical burn	0.7 ^b	0.5 ^{ab}	1.3 ^c	0.3 ^a	1.7 ^d	0.3
<i>Basic tastes</i>						
Salt	6.1 ^a	6.2 ^a	7.0 ^b	7.9 ^b	7.2 ^b	0.3
Sour	1.9 ^a	2.1 ^{ab}	2.5 ^c	2.3 ^{bc}	2.5 ^c	0.2
Bitter	1.3 ^a	1.6 ^a	2.1 ^b	1.6 ^a	2.0 ^b	0.1
Sweet	1.3 ^{ab}	1.3 ^a	1.7 ^{bc}	1.5 ^{ab}	1.9 ^c	0.2
<i>Textures</i>						
Springiness	3.1 ^a	4.9 ^b	5.1 ^{bc}	5.5 ^{cd}	5.8 ^d	0.2
Juiciness	4.1 ^a	4.8 ^b	5.1 ^b	4.7 ^b	4.7 ^b	0.2
Hardness	5.4	5.8	5.3	5.4	5.4	0.2
Cohesiveness	5.9 ^a	6.5 ^b	6.3 ^{ab}	6.0 ^a	6.2 ^{ab}	0.2
Denseness	6.6 ^a	6.6 ^a	6.1 ^a	6.6 ^a	7.4 ^b	0.2

^{abcd} Least squares means in the same row without a common superscript letter differ ($P < 0.05$).

^e Based on the 0–15 Spectrum™ Universal Intensity scale (0 = absence of flavor; 15 = extremely intense flavor).

^f Based on the 0–15 Spectrum™ Universal Intensity scale (0 = not springy, dry, soft, crumbles, airy; 15 = very springy, juicy, hard, defined particle size, dense).

^g SEM: standard error of the mean.

Table 9

Least squares means of descriptive attribute sensory panel scores for color^d, grey color^e, off color^f, and iridescence^g of boneless roast beef stored under refrigerated condition

Storage week	Color	Grey color	Off color	Iridescence
0	4.5 ^a	2.5 ^c	1.1 ^a	2.8
2	5.8 ^b	1.3 ^a	1.4 ^{bc}	2.6
4	5.6 ^b	2.3 ^{bc}	1.9 ^{ab}	2.5
6	5.6 ^b	1.7 ^{ab}	1.4 ^{ab}	2.5
8	5.3 ^b	1.9 ^{abc}	2.1 ^c	2.0
10	5.7 ^b	2.5 ^c	1.9 ^{bc}	2.2
SEM ^h	0.3	0.3	0.3	0.2

^{abc} Least squares means in the same column without a common superscript letter differ ($P < 0.05$).

^d Based on a 8-point intensity scale (1 = grey; 8 = dark reddish pink).

^e Based on a 6-point intensity scale (1/0% = no surface grey color; 6 = total/100% surface grey color).

^f Based on a 6-point intensity scale (1/0% = no surface brown color; 6 = total/100% surface brown color).

^g Based on a 6-point intensity scale (1 = none, 0%; 6 = very strong, 100%).

^h SEM: standard error of the mean.

ged from 0.3–2.7 units. “Salt” was the dominant taste, with a lower perceived score ($P < 0.05$) during the first two wk of storage. Sour and bitter tastes intensified after 4 wk, but these tastes were again barely detectable. Sweetness was higher only on wk 8. After 2-wk storage, springiness and juiciness scores increased ($P < 0.05$) with little to no change reported over subsequent storage. These results indicated that the largest effects due to storage on sensory panel scores were slight increases in the beef/brothy aromatic and springiness after 2-wk storage, with other changes being minimal. Subjective color scores of roast beef slices over the 10 wk storage period were inconclusive with no definitive trends attributed to storage (Table 9).

4. Conclusions

Results of this study indicated that FP or DP could be incorporated into a brine-injected cooked roast beef product at a level of 2.5% without detrimental effects and with the potential benefit of reducing lipid oxidation and WOF. All plum treatments, except 5% PP, caused the roast beef cuts to be slightly darker, while FP reduced redness and DP increased redness slightly when measured by color space values. Plum treatments also had minimal effects on the tenderness of beef roasts as measured by Allo-Kramer shear. Sensory panel evaluation of sliced roast beef noted only that the 2.5% FP was less red than the other treatments. Sensory panel scores indicated a slight plum/prune aroma for FP and DP treated samples, but not the PP. Use of PP is not recommended as a brine ingredient due to reduced product yield.

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